

From the matrix to neuroimage

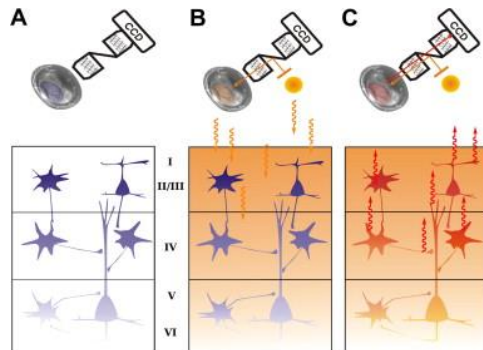
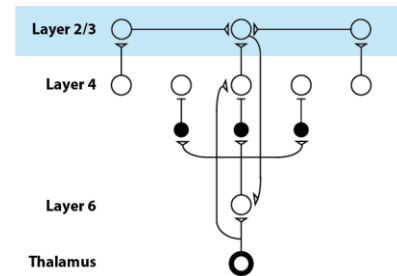
Nicolas Catz
CRPN, AIX-MARSEILLE UNIVERSITÉ



Imaging a cortical map/ Example : Extrinsic Optical imaging (VSDI)

Optical imaging : Voltage Sensitive Dye imaging (Layer 2/3) / VSDI *Mostly supralimninar*

Visualization in real time, of the cortical activity of large neuronal population with high spatial resolution (50 μm / 100*100 pixels) and high temporal resolution (3 ms)



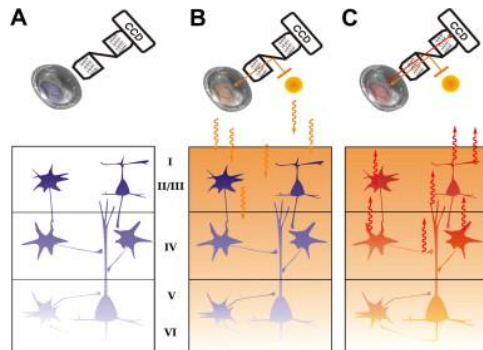
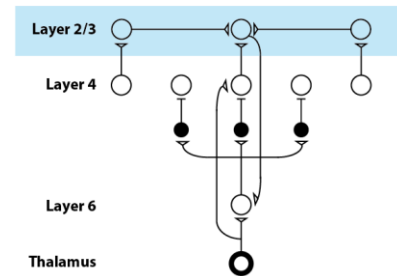
Fluorescent dyes bind to the external surface of membranes of all cell types and act as molecular transducers that linearly transform changes in membrane potential (**mainly synaptic inputs**) into optical signals

Chemla & Chavanne, 2010

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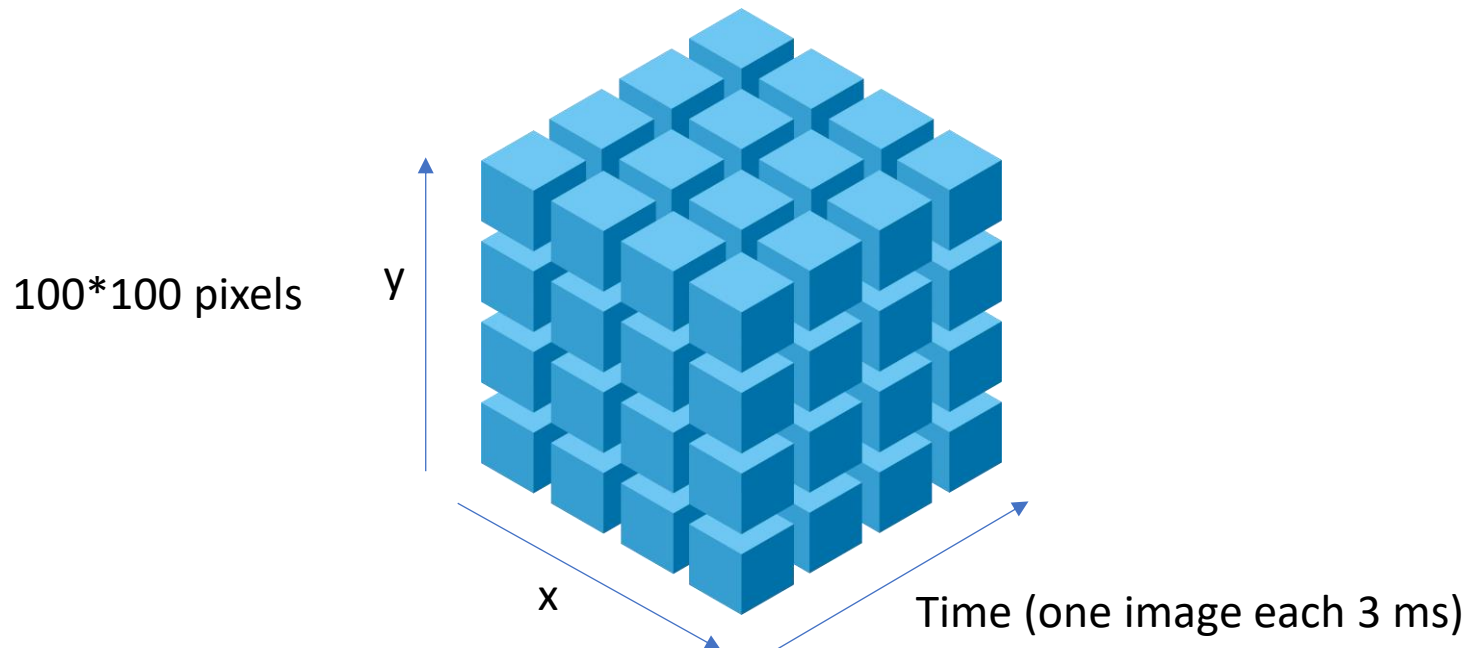


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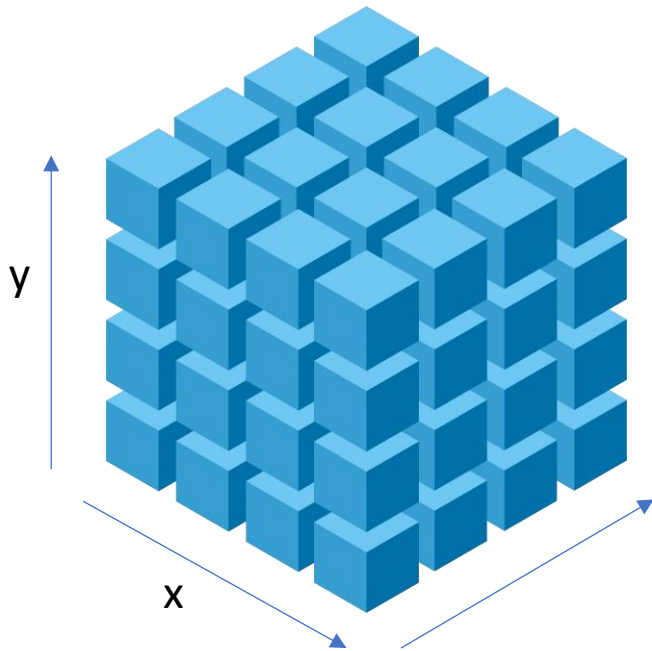
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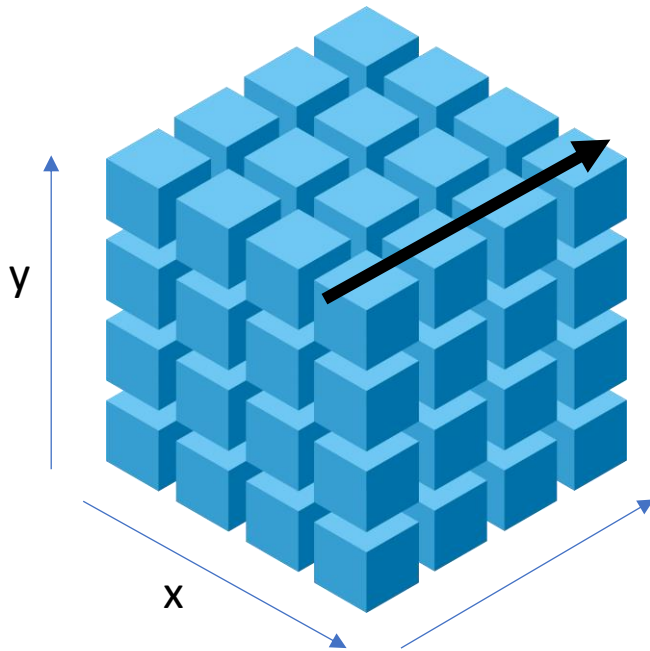
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Each image, one after the other

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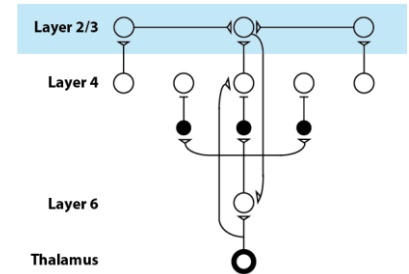
Each image, one after the other

OR

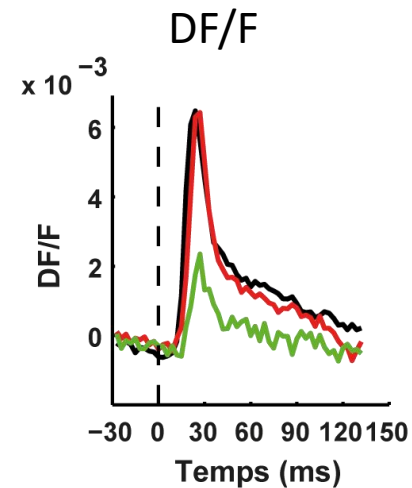
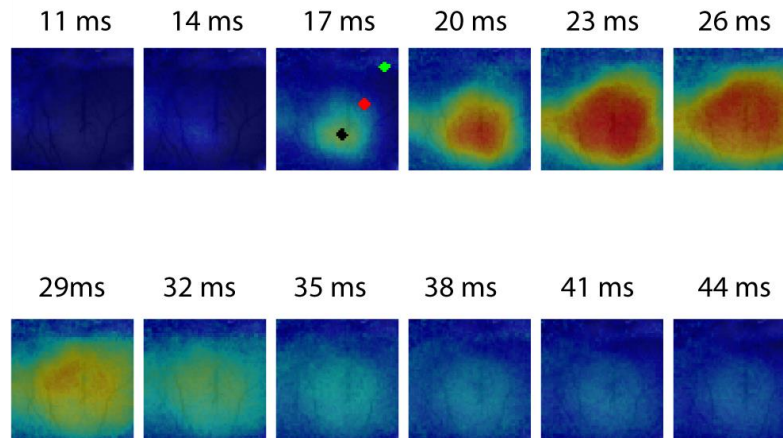
One pixel over time !

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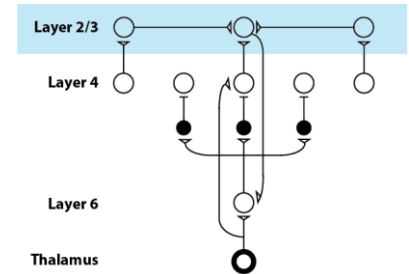


Activation maps

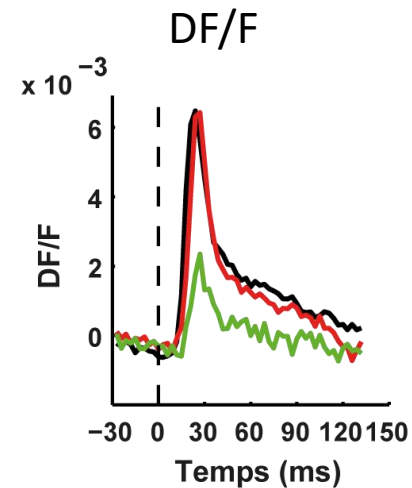
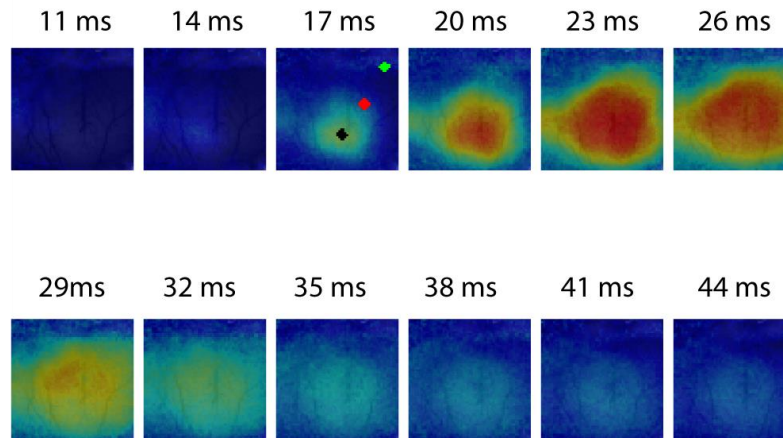


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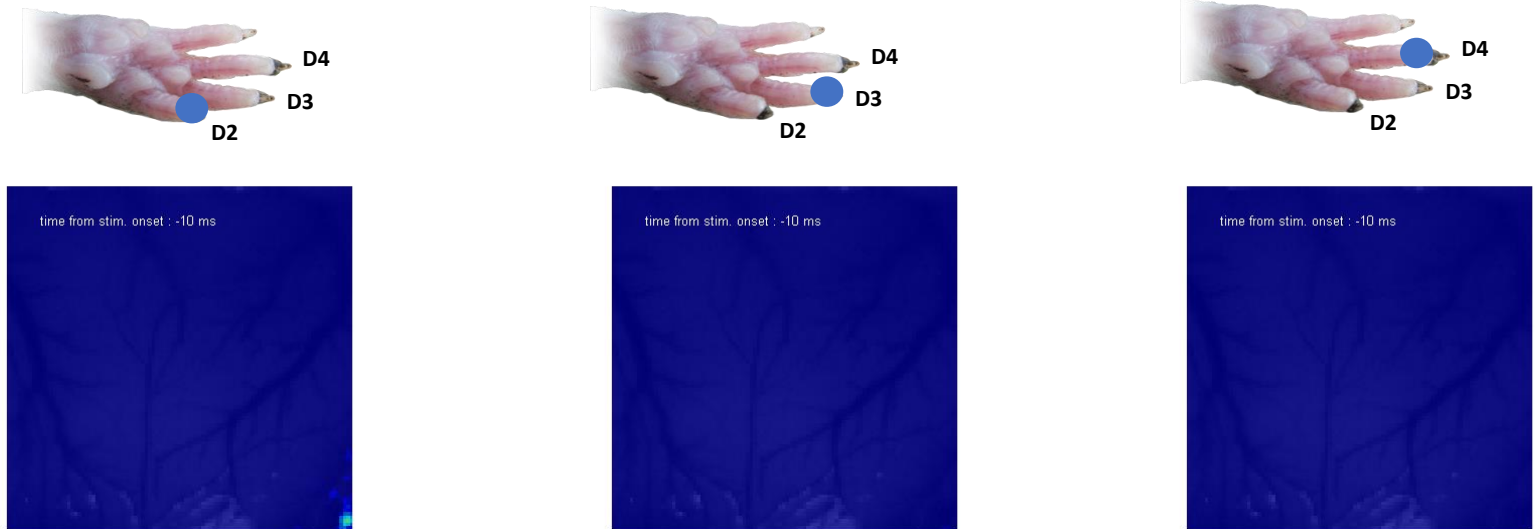


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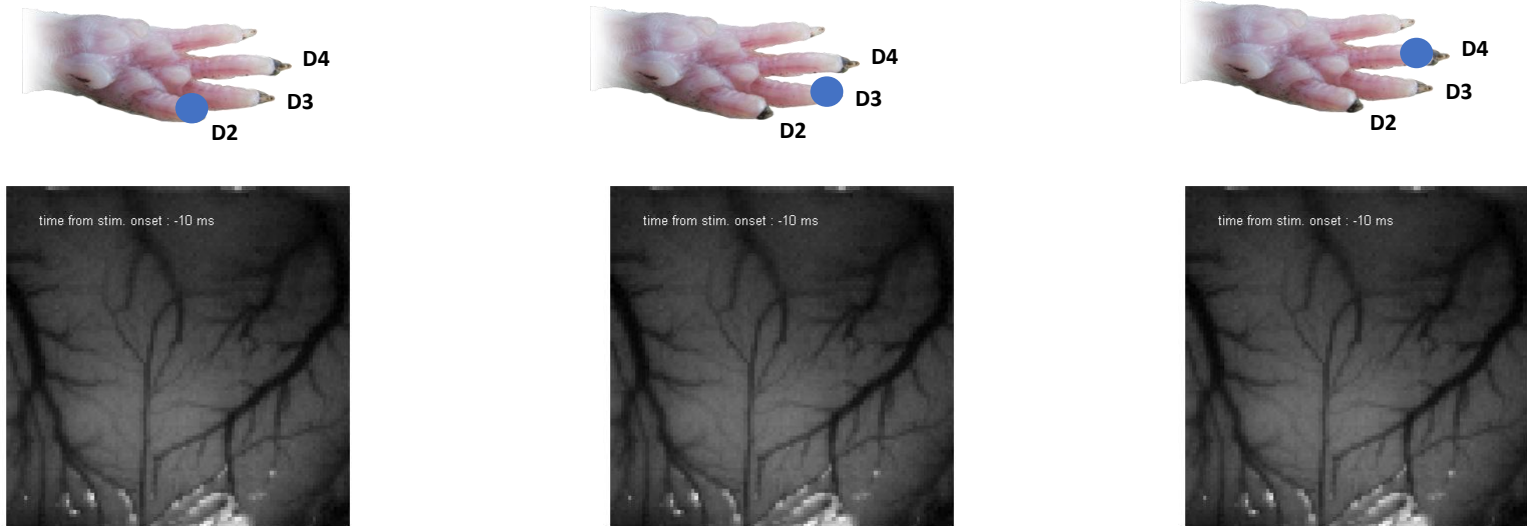
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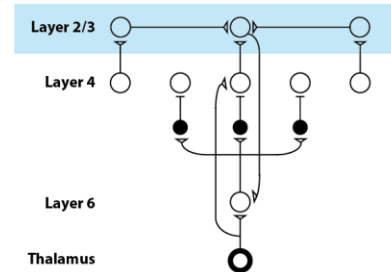
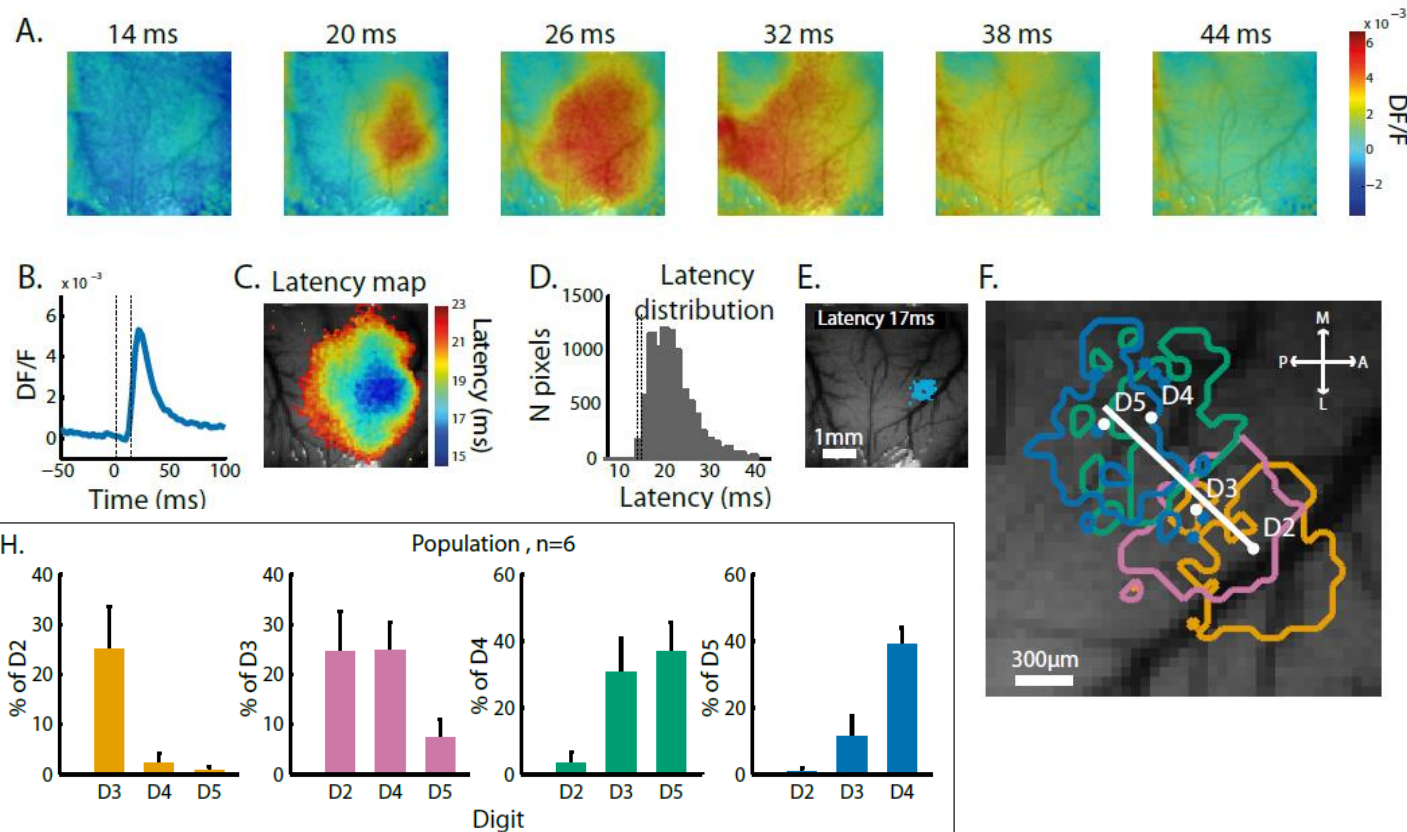
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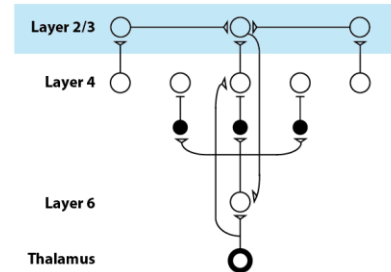
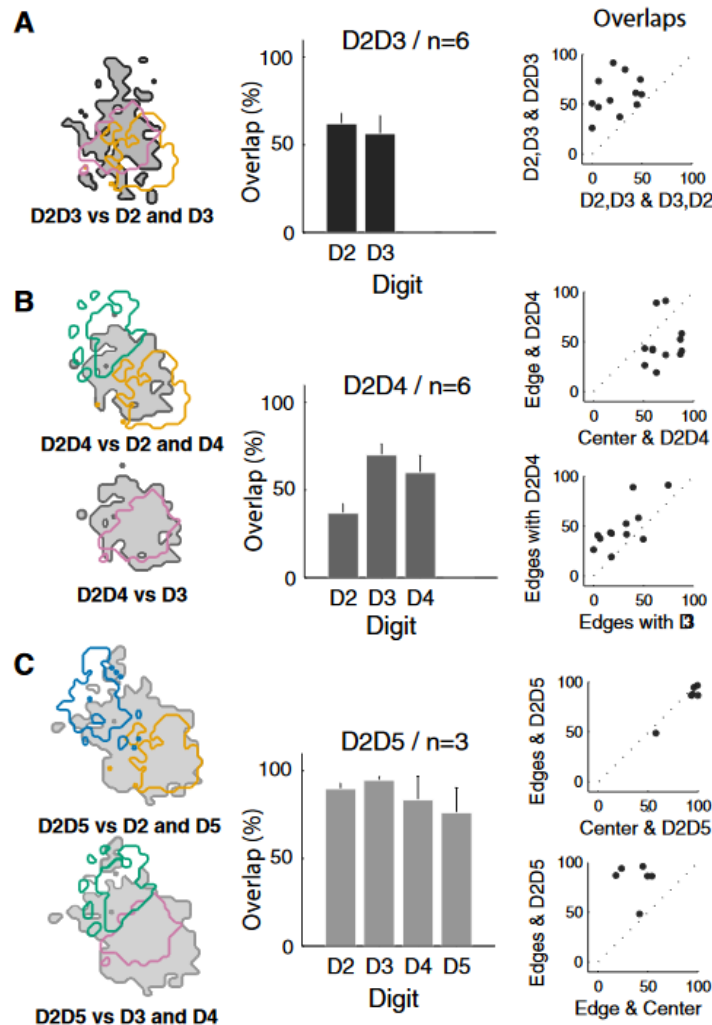
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